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Research and Applications

Med.ai ASK: an agentic system for biomedical question answering

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Abstract

Objective: **Intelligent agent-driven research co-pilots, leveraging advances in generative AI, are transforming how scientists access biomedical knowledge. This paper presents Med.ai ASK, an agentic question-answering system designed to address biomedical inquiries through dynamic retrieval augmentation and tool-driven reasoning. We aim to develop a system capable of parsing the nuance in biomedical scientists' research questions to provide reliable, grounded responses that are more accurate than other generative AI solutions.**

Materials and Methods: **We adopt the ReAct framework's tool-calling architecture and leverage atomic reasoning from Self-Discover to build Med.ai ASK. It selectively queries multiple biomedical knowledge bases and employs map-reduce tools for vector database retrieval, alongside external API and NER tool integration. We ingested 44 million biomedical documents from diverse sources. The agent is evaluated on a range of biomedical question-answering datasets.**

Results: **Human evaluation on an internal dataset shows strong performance and stability. Ratings from a large language model are aligned with human assessments, supporting its use in further experiments. Automatic evaluations indicate superior performance in long-form answers regarding accuracy, faithfulness, factuality, and reduced hallucinations. For short-form and multiple-choice answers, performance is competitive with state-of-the-art systems. The agent's detailed answers are more interpretable than other systems attributed to its agentic design. The agent effectively selects tools based on question type and is deployed in a production-level chat platform with over 1600 users and 25 000 answered questions.**

Conclusion: **Med.ai ASK dynamically orchestrates biomedical information retrieval tools to deliver robust interpretative, accurate, and factual answers, which is crucial in the biomedical domain.**

Key words: biomedical question answering; large language model; retrieval-augmented generation; agentic AI; reasoning.

Background and significance

The rapid advancement of large language models (LLMs) has ushered in a new era for natural language understanding and question answering. However, challenges remain in specialised biomedical domains where accuracy, groundedness and interpretability are paramount. Traditional end-to-end LLM approaches, although impressive in many respects, often lack the mechanisms required for dynamic incorporation and verification of domain-specific knowledge. The retrieval-augmented generation (RAG) technique[1] helps overcome these limitations. In biomedical contexts, the abundance of literature (eg, PubMed), regulatory documents (eg, FDA Drug Labels), and research grants (eg, NIH grants) makes RAG particularly valuable for supporting factually correct and contextually rich responses. Motivated by this, we propose Med.ai ASK—an agent that leverages a tool-oriented design to integrate multiple biomedical data sources.

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A RAG-based system usually has 2 steps: a retrieval step that returns relevant documents of an input query, and a generation step that uses those documents as context to generate answers. Such systems have been implemented for question-answering tasks: in the general domain, eg, RAG-HAT[2] and HippoRAG;[3] in the biomedical domain, eg, DMIIP,[4] UR-IW[5] and MiBi,[6] including participants in the BioASQ shared task 2024[7] and in the clinical domain.[8] While these systems show improved performance, they exhibit 2 limitations. Firstly, they are designed and finetuned for specific datasets, leading to out-of-distribution risks and increased scaling demands from repeated fine-tuning as new datasets arise. Secondly, their retrieval phase uses only 1 tool, reducing their adaptability when several data sources could help. In contrast, our proposed agent addresses both constraints by incorporating a diverse set of tools, enhancing its flexibility and enabling automatic tool selection based on the question.

Agentic AI systems enhance RAG frameworks by adding autonomous decision-making and dynamic reasoning to integrate diverse data streams in real time.[9,10] Models used by such systems can follow instruction-based reasoning and logic frameworks to address specific tasks, like ReAct,[11] Chain-of-Thought,[12] or Tree-of-Thoughts.[13] Those systems require an orchestration layer that governs how an agent takes in information, performs some internal reasoning, and uses that reasoning to inform its next action or decision.[9] Another line of Agentic AI systems is to support researchers with tasks like literature reviews, experimentation, and report writing, such as Google AI co-scientist,[14] Sakana AI scientist,[15] and ResearchAgent.[16] Med.ai ASK adopts an agentic design tailored for the question answering (QA) task, enabling autonomous selection and invocation of appropriate tools, and determining when tool use should continue or stop. Its design remains intentionally simpler than the broader scientific agents, focusing exclusively on QA performance.

Notably, ASK is designed to be dataset-agnostic and does not rely on task-specific fine-tuning. This enables seamless deployment across diverse biomedical QA benchmarks without retraining or architectural changes. By avoiding dataset-specific dependencies and supporting multiple tools, ASK directly addresses the scalability and adaptability limitations of existing RAG systems. Combined with our comprehensive benchmarking across multiple datasets and evaluation metrics, our work offers a robust, ready-to-use solution for researchers and practitioners seeking reliable performance in real-world biomedical QA.

Objective

Our objective is to develop a research assistant agent for diverse stakeholders in a large pharmaceutical organization. Built on the core ideas ReAct[11] and Self-Discover frameworks,[17] our system incorporates multiple resources, including PubMed abstracts and full-text papers, Elsevier e-books, FDA Drug Labels, the NIH Grant Database, ClinicalTrials.gov, UniProt and a biomedical named entity recognition tool. Unlike the original Self-Discover framework, which follows a 3-step process of *select*, *adapt*, and *implement*, our design streamlines operations by eliminating the *adapt* phase. During the *select* step, all necessary parameters for the subsequent tool execution phase are preconfigured, thereby bypassing the need for real-time adaptation. The agent is further enhanced by an integrated *memory module*[18] that records session history, providing continuity and context for multi-turn conversations.

Finally, we aim to comprehensively benchmark the proposed system using various evaluation metrics across a diverse range of biomedical datasets. We evaluated performance on 4 datasets: BioASQ-12B (released in 2024),[19] LitQA,[20] MedQA,[21] and GeneTuring.[22] Additionally, manual evaluations were performed on an in-house dataset containing 20 questions. On BioASQ-12B, ASK demonstrated higher factuality, accuracy, faithfulness, context precision and recall, and lower hallucination rate relative to our baselines and MiBi.[6] For LitQA, our system achieved superior accuracy and precision compared to current state-of-the-art (SOTA) models. In MedQA, the system performed competitively with Med-PaLM2,[23] although its performance did not match that of Med-Gemini.[24] Similarly, while ASK outperformed the 3 baseline models on the GeneTuring dataset, the performance remained below that of Med-Gemini. These findings indicate that our system not only maintains competitive performance on certain benchmarks but also delivers robust coverage across a wide spectrum of biomedical datasets.

Materials and methods

User facing application

We develop Med.ai ASK by following the common architecture of a front end connected with a proxy backend via a web socket as shown in Figure 1. The proxy backend communicates with our agent backend service via REST API. All ASK services are deployed and orchestrated on a Kubernetes cluster hosted in the cloud, capable of handling enterprise-wide traffic at a large organization. The production-level platform can collect user rating of the answers both as a binary thumbs up/down and as free-text feedback. More information about the core backend and its tools will be detailed in the following subsections.

As depicted in Figure 2a, upon receiving a question, the chatbot generates a response accompanied by corresponding citations with clickable links, thereby facilitating users' verification of information. During the retrieval process, the agent displays its intermediate reasoning—such as the tools invoked and their respective outputs—thereby elucidating the reasoning and tool selection steps undertaken to derive an answer. This contributes to the interpretability and explainability of the agent framework, as illustrated in Figure 2b. Furthermore, leveraging a memory schema, the platform archives all chat sessions, allowing users to reference previous interactions; in each session, prior questions and answers are used as contextual information for subsequent turns, and users may resume sessions later.

The question-answering agent

Inspired by the ReAct agent's[11] tool calling architecture and the Self-Discover framework,[17] we replace reasoning components with tool descriptions used for the QA task. As depicted in Figure 1, based on an input question and the provided tool descriptions, the "Tool orchestration" component decides which tools will be invoked using an LLM. Then the "Tool node" will

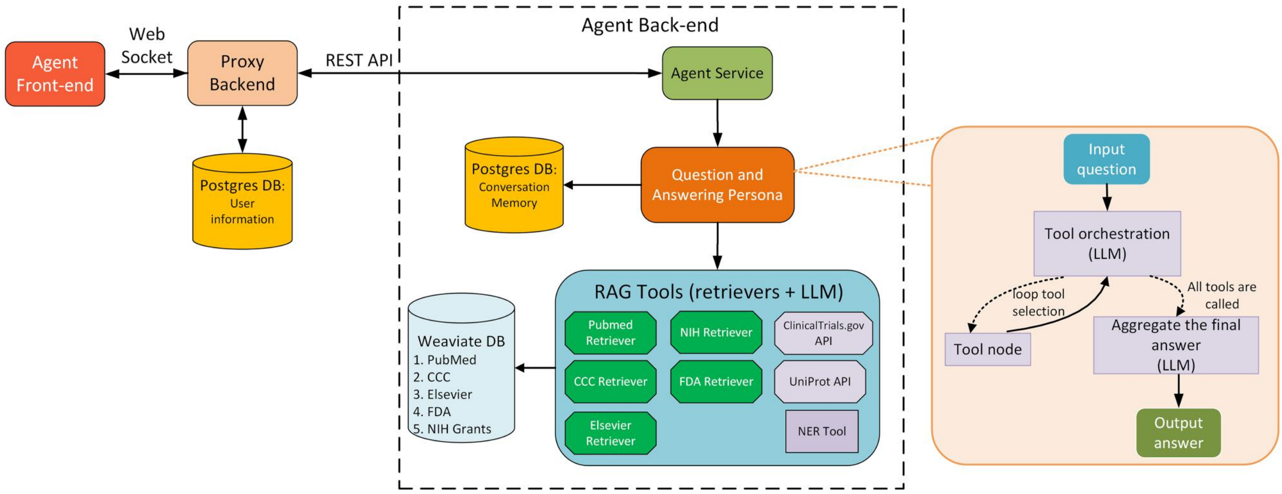


Figure 1. An overview of the agent architecture. The core of our backend is the agentic question-answering persona that interacts with a biomedical tool suite and a database for memory retrieval. In our framework, we leverage LLMs in 3 places: the tool orchestration, the aggregate (summary) node, and RAG tools for our retrievers.

iteratively invoke the selected tools to generate candidate answers. Based on the input question, the agent formulates queries to collect candidate answers from each tool. After the tools return their outputs, the agent synthesizes a final answer by aggregating and reasoning over the collected responses. If the agent cannot find any relevant answer from the knowledge bases, it will output "No answer could be found" rather than relying on the world knowledge of the LLMs, which is overly risky in the high-stakes biomedical domain. The agent prompt is detailed in Appendix A1. The whole workflow is implemented using Langgraph (https://www.langchain.com/langgraph).

Knowledge bases and tools

Document indexing and retriever

We index documents using Weaviate, hosted in a virtual private cloud (VPC). Short documents (eg, PubMed abstracts) are stored as single chunks, while longer documents are split into smaller chunks of 8000 characters with an overlapping of 1000 ones. Each chunk is embedded using the m-GTE model.[25] We internally ingest 5 textual knowledge bases:

- PubMed: 38 million citations (https://pubmed.ncbi.nlm.nih.gov/) for biomedical literature from MEDLINE and life science journals.
- CCC OpenAccess: A licensed collection of 5 million full-text biomedical papers.
- Elsevier ebooks: A licensed collection of 20 000 ebooks about biology.
- NIH Grant Database: A collection of 1 million research grants submitted to National Institutes of Health (NIH).
- FDA Drug Labels: A collection of over 150 000 labelling documents for human prescription drug, nonprescription drugs, and labelling for other products (https://www.fda.gov/science-research/bioinformatics-tools/fdalabel-full-text-search-drug-product-labeling\#Database\%20Access).

At the retrieving step, we leverage the weaviate python library to retrieve relevant documents for each input query.

In addition to the internal retrievers, we also employ 2 API-calling tools:

- ClinicalTrials.gov API (https://clinicaltrials.gov/data-api/api) to access clinical research studies and information about their results.
- UniProt API (https://www.ebi.ac.uk/proteins/api/doc/) to retrieve information about proteins.

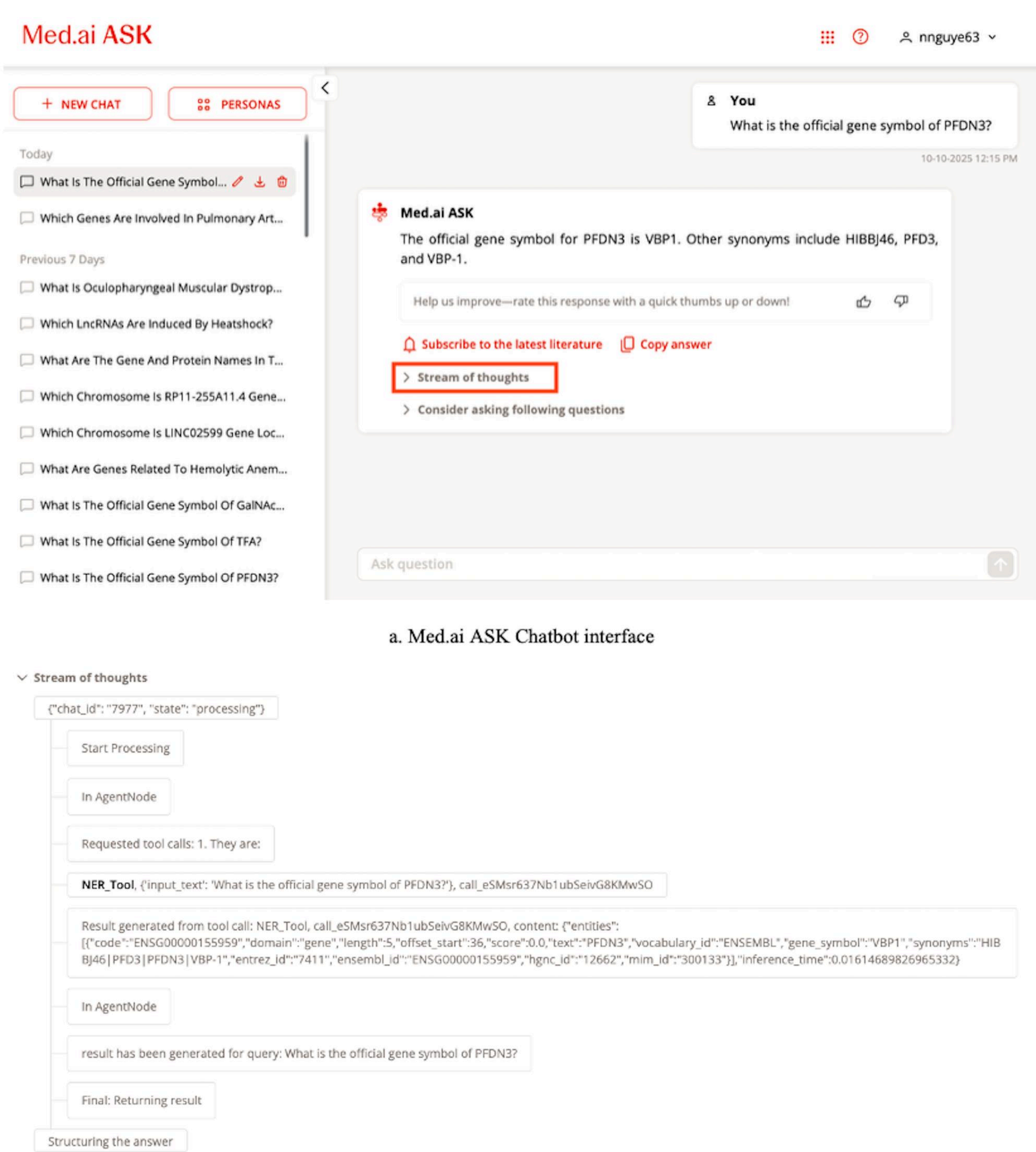
Answer generation

Based on each retriever, we implement a QA RAG tool using a map-reduce QA algorithm to generate answers. In the first (map) step, each relevant documents returned by the corresponding retriever is fed into an LLM with a prompt (see Appendix A2) instructing to answer the question to produce document-level answers. In the reduce step, the document-level answers are collectively summarized to answer the question, while appending document IDs (such as PubMed ID) of the summarized documents to each statement in-line to ensure interpretability of our system (see Appendix A3 for the reduce prompt). This builds upon Langchain map-reduce summarization implementation.

We also implement a named entity recognition (NER) tool by wrapping the Kazu framework[26] and a bespoke gene synonym expansion step using NCBI gene data.[27] This NER tool can identify and normalize to controlled vocabularies of biomedical entities in 12 different categories.

Memory

We implement a simple schema for our agent's memory. Specifically, at chat session level, we save every question and its answer in a Postgres DB. When a user asks a new question in a chat session, information of previous questions will be retrieved and attached as context of the new question.



a. Med.ai ASK Chatbot interface

b. Med.ai ASK Stream of thoughts

Figure 2. (a) The Med.ai ASK user interface. When clicking on "Stream of thoughts," a user will see the information depicted in (b). (b) The Agent's stream of thoughts. Every tool selection and tool's information are shown to users so that they know what steps the Agent uses to derive an answer.

Datasets

We evaluate our ASK agent on 5 different QA datasets covering long-form answers, span-based answers and multiple-choice answers.

In-House Dataset has 20 user-submitted questions with negative feedback from previous Med.ai ASK releases. Each question, reflecting real-world scenarios that sometimes extend beyond fact-finding and information retrieval, has a reference answer from 1 domain expert that was checked by another. Their templates are shown in Appendix B.

BioASQ-12B (2024) was used in the BioASQ-12B shared task of 2024.[19] For our evaluation, we used "ideal" long-form answers (as opposed to "exact" shorter answers available in the dataset), that were manually constructed based on relevant PubMed abstracts. The dataset has 5049, and 340 questions for training and testing sets, respectively.

For a granular analysis of performance, each BioASQ-12B testing question was assigned up to 3 out of 37 predetermined topics within the biology domain by an LLM; the resulting topic labels were finally validated by the domain experts. Most of the questions fall into Disease & Symptoms, Treatments, Clinical Medicine, Oncology, and Pharmacology topics (see Appendix C for more details).

LitQA-v2 was specifically designed to evaluate RAG systems.[20] It has 199 multiple-choice questions with unambiguous answers that are generally only attested to once in the scientific literature and are not answerable by information in abstracts (ie, they are usually derived from findings reported in relatively recent scientific papers).

MedQA is a Multiple-Choice Question Answering (MCQA) dataset, with questions drawn from the medical licensing examinations in the United States, mainland China, and Taiwan.[21] In this work, we only experiment with the United States subset, ie, USMLE.

GeneTuring includes 600 open/close-ended QA pairs to test the performance of LLM models in 12 different categories about genomic knowledge.[22] All questions have a span-based answer, ie, each question requires an entity as an answer.

We summarized the characteristics of all selected datasets in Table 1.

Evaluation metrics

Conventional accuracy

For each question, we do exact matching on the ground truth and the generated answer.

LLM-based accuracy

Inspired by the LLM-as-a-judge approach,[28] we prompt an LLM to compare the generated long-form answers with the ground truths and ask it to provide a score between 0 and 1 (see Appendix A4 for the prompt we used). We calibrate LLM accuracy by comparing it to human ratings as described below.

OLAPH metrics

We also use 4 OLAPH metrics[29] for long-form answers: Word composition, Semantic similarity, Factuality, and Hallucination. For factuality and hallucination scores, instead of leveraging an LLM to generate nice-to-have and must-have statements, we directly break the ground truths into such statements.

RAGAs metrics

We employ 3 RAGAs metrics:[30] faithfulness, context precision, and context recall to assess how accurately and effectively the retrieved documents support the generated answers. These metrics provide a quantitative measure of the reliability and grounding of our system's outputs.

Human evaluation metrics

We introduce 2 criteria to manually evaluate generated answers: accuracy and stability. For accuracy, an expert will rate 1 if the generated answer is accurate and matched to the ground truth; otherwise, they will rate 0. Meanwhile, the stability score is calculated based on the similarity among answers. For example, given n generated answers, if all answers are similar, the stability is 100%. If there are m similar answers, the stability is $\frac{m}{n}$. It is noted that in this study, we provided the experts with 3 different generated answers for a subset of 20 user-submitted questions extracted from our in-house dataset.

Inspired by RAGAs, we define 2 criteria to manually evaluate the cited documents in generated answers:

- Relevance: measures whether the cited documents include relevant information to answer the question.
- Faithfulness: assesses whether the cited documents genuinely support the content of the generated answer.

In both cases, the score is the number of correct documents divided by the total cited documents.

Evaluation settings

We compare our agent with 3 baselines: LLaMa3.1-8B-Instruct,[31] GPT-o3, and GPT4.1; each is directly prompted without retrieval information. On BioASQ-12B, we also compare with MiBi,[6] a top-performing system from the 2024 shared task. We first adapted their pipeline by replacing Elasticsearch with our PubMed retriever and to ensure the fairness, we upgraded the base LLM from GPT-3.5-turbo to GPT-4.1. For other datasets, we include comparisons with available SOTA systems, though some metrics are unavailable due to limited coverage.

Evaluation metrics vary by dataset. For BioASQ-12B, we report LLM-based accuracy, OLAPH, and RAGAS metrics. Since RAGAs are specific to RAG systems, they are reported only for MiBi and ASK. Traditional accuracy is used for MedQA and GeneTuring, while accuracy and precision are reported for LitQA-v2, following Laurent et al.[20] GPT-4.1 is used for LLM-based accuracy, OLAPH and RAGAS scoring.

Table 1. An overview of the selected datasets.

Dataset	Answer types	Train	Dev	Test
BioASQ-12B	Long-form	5049	503	340
LitQA v2	Multiple choice	NA	NA	199
MedQA (USMLE)	Multiple choice	10 178	1272	1273
GeneTuring	Span-based	NA	NA	600
In-house	Long-form	NA	NA	20

We evaluate our systems using the testing set only.

Med.ai ASK uses GPT-4.1 as its foundational LLM and the optimal RAG parameters (see Appendix D for selection details). It does not rely on any training data.

All results are averaged over 3 independent runs on the test sets.

Results

Human evaluation

First, we compare the accuracy rated by domain experts and LLM on our in-house dataset to determine validity of applying LLM-as-a-judge. Overall, the agent's average performance was evaluated at 84% ±2.5 by domain experts and 83.7% ±1.9 by an LLM. The average Krippendorf alpha was 0.523 ± 0.029, and the pairwise F_1 was 0.755 ± 0.015, indicating moderate inter-annotator agreement.

Qualitative analysis revealed discrepancies between the 2 evaluation approaches. For instance, in response to the question, "What is the 5-year mortality rate of [disease]?," the ground truth reported rates by age group, while the agent's answer provided rates by geographical region. As a result, the LLM assigned an accuracy score of 50% due to mismatched numerical values, whereas the human annotator awarded a score of 100% based on the overall quality of the response. Another disagreement occurred with a question about the "most cited papers," which requires counting citations rather than simply retrieving articles. In this case, the agent identified 1 correct paper and supplied information about additional papers. The LLM rater gave an accuracy score of 100%, while the domain expert rated it at 30%.

In terms of citation evaluation, the average scores are 89.1%± 2.4 of relevance and 82%±3.5 of faithfulness on the in-house dataset, indicating strong citation reliability of the agent.

Automatic evaluation

BioASQ-12B

We depicted performance of all systems on the BioASQ-12B testing set in Figure 3. The confidence intervals were calculated by using a bootstrap technique across all questions over 3 runs. Among the evaluation metrics, Med.ai ASK outperformed both the baseline systems and MiBi on LLM-based accuracy, Factuality, and Hallucination. The agent did not achieve better scores on the Word Composition and Semantic Similarity metrics; however, we maintain that these measures are not critical for assessing response quality in this context.

Figure 4 presents hallucination scores across topics, where lower values indicate better performance. ASK outperformed all 3 baselines on most topics, except Epigenetics, and showed slightly better performance to MiBi overall. Notably, the agent produced no hallucinations in 6 topics: Virology, Vaccines, Physiology, Pharmacovigilance & Adverse Effects, Cell Biology, and Biochemistry; MiBi matched this except for Pharmacovigilance & Adverse Effects. Stratified results for other metrics are provided in Appendix E.

Evaluation results on cited documents in Figure 5 indicate that ASK consistently outperformed MiBi over the 3 metrics. While MiBi performs moderately well, its scores remain notably lower, especially in context precision and recall.

LitQA-v2, MedQA and GeneTuring

As shown in Table 2, on the LitQA-v2 dataset ASK outperformed all baselines and current SOTA systems on both accuracy and precision. On the MedQA dataset, our agent performed better than LLaMa3.1 but only achieved competitive results compared to Med-PALM2,[23] while falling short of the current SOTA systems (GPT-o3 followed by Med-Gemini[24]) For the GeneTuring dataset, our agent demonstrated improvement over the 3 baselines, though it did not match the performance of Med-Gemini, the current leader on this benchmark.

Qualitative analysis on multi-choice questions revealed that where questions lack essential details, Med.ai ASK can discern and explain the deficiency, as illustrated in Appendix G1.

Tool contribution

Ablation study

To evaluate the role of each tool, we incrementally integrated them into our agent and assessed its performance on the development set of BioASQ-12B. For their individual performance, please refer to Appendix D. In Table 3, we report the accuracy as well as *P*-values when comparing (1) the current run to the previous one and (2) the current run to PubMed. Gradually adding tools into the tool collection led to improved average accuracy. The most pronounced improvement was observed when PubMed was first added, with a significant difference between the PubMed-enabled and no-tool configuration. Adding more tools on top of PubMed resulted in less marked differences. Nevertheless, compared to the PubMed-only model, further enhancements were significant in most cases.

Tool usage

We also calculated the frequency of tool usage for answering questions on different datasets and report the results in Table 4. While PubMed was used the most in BioASQ-12B, LitQA-v2, and MedQA datasets, for GeneTuring the agent used the NER tool the most. When looking at each tool individually, CCC has the highest usage in the LitQA dataset, Elsevier in BioASQ, and UniProt in GeneTuring. Meanwhile, NIHGrant and CTGov have the least usage.

User engagement

To date, the Med.ai ASK platform has served more than 1600 unique users who have collectively submitted more than 25 000 inquiries as depicted in Figure 6. We have received outstanding feedback from our stakeholders such as "We've rapidly honed in on key literature to drive risk assessments with Med.ai ASK, saving hours in FTE time!," "Med.ai ASK is delivering really consistent rigor to our literature searches as we derive risk assessments," and "Med.ai ASK was instrumental in helping me structure my thinking and locating critical literature." So far, 6 internal teams have expressed interest in potential integrations between Med.ai ASK and their proprietary data or solutions.

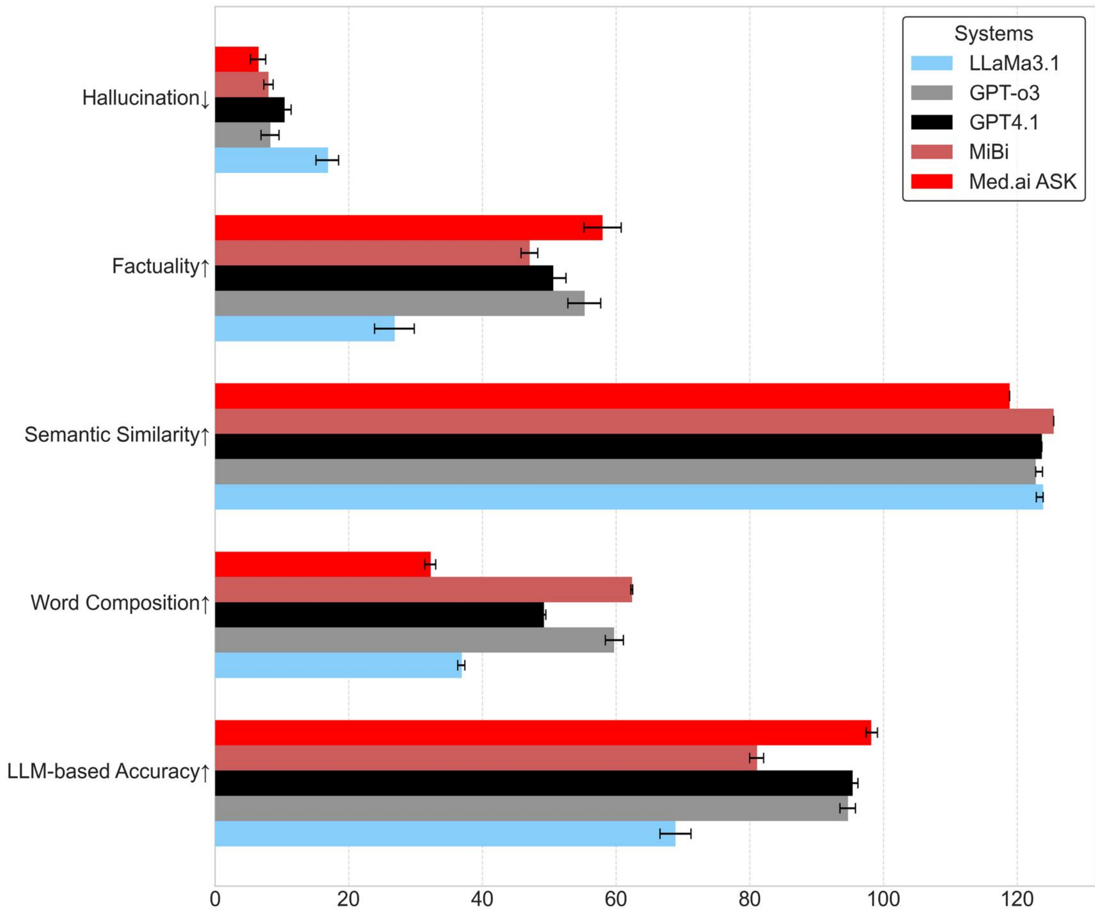


Figure 3. Performance on BioASQ-12B testing set on 5 different evaluation metrics. For the Hallucination metric, a lower value indicates better performance, whereas for the other metrics, higher values indicate better performance. It is noted that each metrics has a different range.

Discussion

First, we evaluated LLM-based accuracy rating of answers by manually comparing it with human annotations. The close correspondence observed (84% by human vs 83.7% by LLM) justified the use of an LLM accuracy rater in subsequent experiments. Some discrepancies between the human and LLM-as-a-judge may be attributed to LLM being stricter, while humans being more lenient to certain arbitrary choices made to construct ground truth answers or LLM predicted answers. We acknowledge that using only 20 questions limits our evaluation. However, each question required manual review across 3 runs, making it resource intensive. We therefore strengthen the assessment with automatic metrics.

Our agent performed better than our baseline and MiBi systems in long-form answers, eg, the BioASQ-12B dataset. It provided more accurate and more factual answers while keeping the hallucination metric as low as possible (cf. Figures 3 and 4). ASK also demonstrates strong ability to retrieve, use, and accurately reflect information from source documents (cf. Figure 5). LitQA-v2 is challenging than MedQA because the number of choices is not fixed in each question. Nevertheless, ASK outperformed SOTA systems in both accuracy and precision scores (cf. Table 2). However, the agent performed worse on the MedQA and GeneTuring datasets. A likely explanation is that answering questions from these datasets requires consulting additional resources beyond literature, eg, medical textbooks for MedQA and the GeneOntology for GeneTuring. Consequently, integrating web search tools into the agent will be our future work.

LLMs are known to frequently generate hallucinations when answering multiple choice questions because the training and evaluation procedures reward guessing over acknowledging uncertainty.[32] To better serve our users, we prioritized factual accuracy, faithfulness and clear citation of sources over improving performance on MCQA datasets. Consequently, answers generated by Med.ai ASK tend to be more verbose than those generated by other systems (cf. Appendix G2). Such increased verbosity stems from the agent's design to enhance transparency by explicitly detailing the intermediate reasoning steps involved in generating an answer. This approach improves evidence citation by incorporating extensive, clickable references, allowing users to verify the sources of information. Additionally, the elaboration of the decision-making process enhances interpretability by showing the specific tools utilized by the agent.

We identified outdated ground truths in BioASQ-12B, particularly for Clinical Trial questions. For example, the ground truth for "What are the primary and secondary endpoints of Pamrevlumab?" cites only 1 endpoint for each category from a 2023 publication, despite the existence of at least 3 trials with

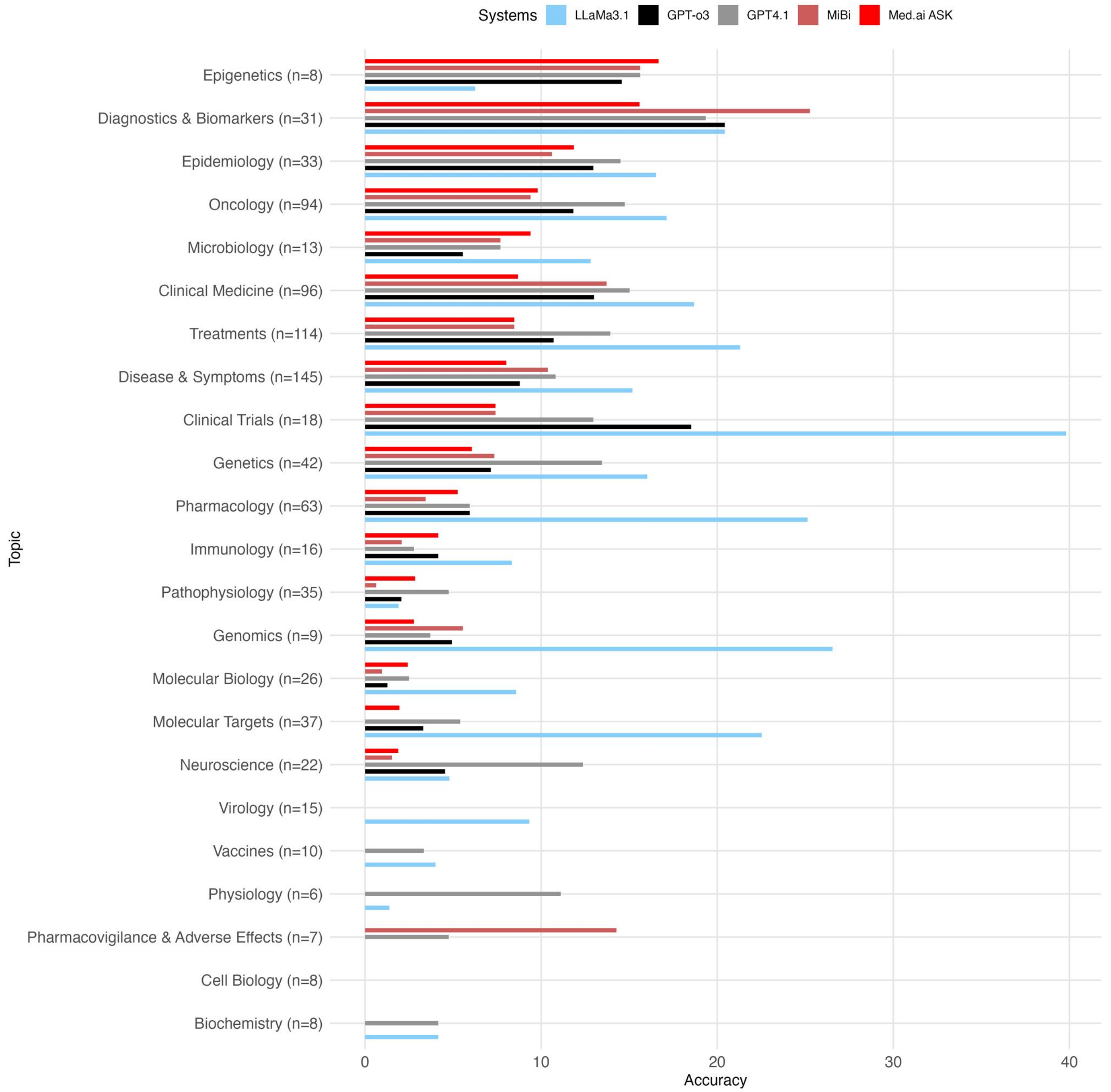


Figure 4. Hallucination scores stratified by topic (with more than 5 questions). Lower scores indicate better performance, with 0 representing no hallucinations. Med.ai ASK outperformed LLaMa3.1, GPT-o3 and GPT4.1 in 18 over 23 depicted topics. ASK is better than MiBi in 6 topics and competitive in the other topics.

distinct indications. Using RAG tools, our agent provided comprehensive and up-to-date answers, referencing sources from 2024 and 2025 (cf. Appendix G2). However, these updated responses can be mistakenly classified as hallucinations when compared to outdated ground truths, making evaluation challenging. A similar issue was observed in the MedQA dataset, where test questions contain obsolete answers. Saab et al.[24] also noted that many questions lack critical details, eg, figures or laboratory results. Although they subsequently revisited the dataset labels, the revised version was not published.

The ablation study indicates that incorporating all available tools does not necessarily improve overall performance (cf. Table 3), potentially due to temporary confusion within the LLM's decision-making process. Nonetheless, leveraging diverse sources is critical for achieving comprehensive coverage across various topics. In Table 4, although the NER tool was less frequently employed in other datasets, it emerged as the primary tool for GeneTuring. For real-world applications like the Med.ai ASK chatbot, the inclusion of sources such as FDA Drug Labels and NIH Grant documents is anticipated to be beneficial.

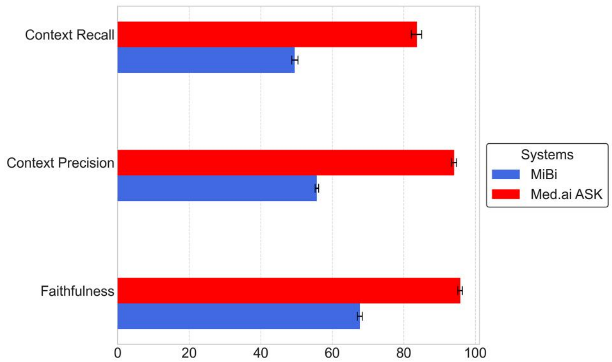


Figure 5. Citation evaluation results on Med.ai ASK and MiBi. ASK consistently outperformed MiBi on faithfulness, context precision and context recall, highlighting Med.ai ASK's stronger ability to retrieve, use, and accurately reflect information from source documents. Topic stratified results for these metrics are provided in Appendix E.

Table 2. Performance on multiple-choice (LitQA-v2, MedQA) and span-based (GeneTuring) datasets.

System	LitQA-v2		MedQA (Accuracy)	GeneTuring (Accuracy)
	Precision	Accuracy		
LLaMa3.1-8B	35.0	28.5	50.8	22.5
GPT-o3	<u>44.3</u>	<u>44.0</u>	**96.1**	35.2
GPT4.1	42.0	41.9	89.1	32.4
Med.ai ASK	**65.1**	**43.9**	86.2	<u>41.8</u>
Claude3-haiku[20]	38.0	27.0	–	–
LLaMa3-70B[20]	38.0	35.0	–	–
Hou et al.[22]	–	–	–	51.2
Med-PaLM2[23]	–	–	86.5	–
Med-Gemini[24]	–	–	<u>91.1</u>	**54.5**

Dash (-) indicates no data available. Our scores are averaged on 3 independent runs. Bolded numbers indicate the best scores; underlined numbers indicate the second-best ones. For a break-down performance on 12 categories in the GeneTuring dataset, please check Appendix F.

Table 3. LLM-based accuracy on the development set of BioASQ-12B when we gradually add tools into our agent.

Tool configuration	Accuracy mean±SE	*P*-value (vs prev.)	*P*-value (vs PubMed)
No tool (GPT4.1)	92.2 ± 0.1		
PubMed	96.1 ± 0.5	.015*	
PubMed + CCC	97.9 ± 0.3	.050	.050
PubMed + CCC + Elsevier	98.1 ± 0.3	.677	.041*
PubMed + CCC + Elsevier + UniProt	98.4 ± 0.4	.578	.027*
PubMed + CCC + Elsevier + UniProt + NIHGrant	98.2 ± 0.2	.647	.044*
PubMed + CCC + Elsevier + UniProt + NIHGrant + CTGov	98.5 ± 0.1	.266	.037*
PubMed + CCC + Elsevier + UniProt + NIHGrant + CTGov + FDA	98.2 ± 0.2	.370	.041*
PubMed + CCC + Elsevier + UniProt + NIHGrant + CTGov + FDA + NER	97.8 ± 0.1	.130	.075

Bolded numbers indicate the best scores. *P*-values were calculated using t-test.

When building the Med.ai ASK platform, we aimed at addressing QA tasks. Nevertheless, its flexible agentic architecture enables it to handle additional functions beyond conventional QA. For instance, it can compare lists of genes or drugs, generate result tables, and automatically filter literature based on specific timeframes. Consequently, users also employ the agent to facilitate comprehensive literature searches.

Conclusion

In this paper, we introduced Med.ai ASK, an agentic system that leverages a tool-oriented design to dynamically integrate multiple biomedical knowledge sources. Our extensive evaluations on the BioASQ-12B, multiple choice QA, span-based QA, and in-house datasets demonstrate that our agent consistently delivers

Table 4. Percentage (%) of tool usage by the agent on different datasets.

Tools	BioASQ-12B	LitQA-v2	MedQA	GeneTuring
PubMed	**28.8**	**44.2**	**46.9**	29.3
CCC	25.2	33.3	27.3	18.0
Elsevier	22.3	11.9	13.9	0.5
UniProt	5.5	2.2	0.9	21.7
NIHGrant	0.1	0	0	0
CTGov	2.0	0	0	0
FDA	6.6	0.1	1.1	0
NER	9.5	8.3	9.9	**30.4**

The percentage usage of each tool was calculated as the number of times the tool was invoked divided by the total number of tool invocations to answer questions in a specific dataset. Those numbers are then averaged via 3 runs. Bolded numbers indicate the highest usage in a dataset.

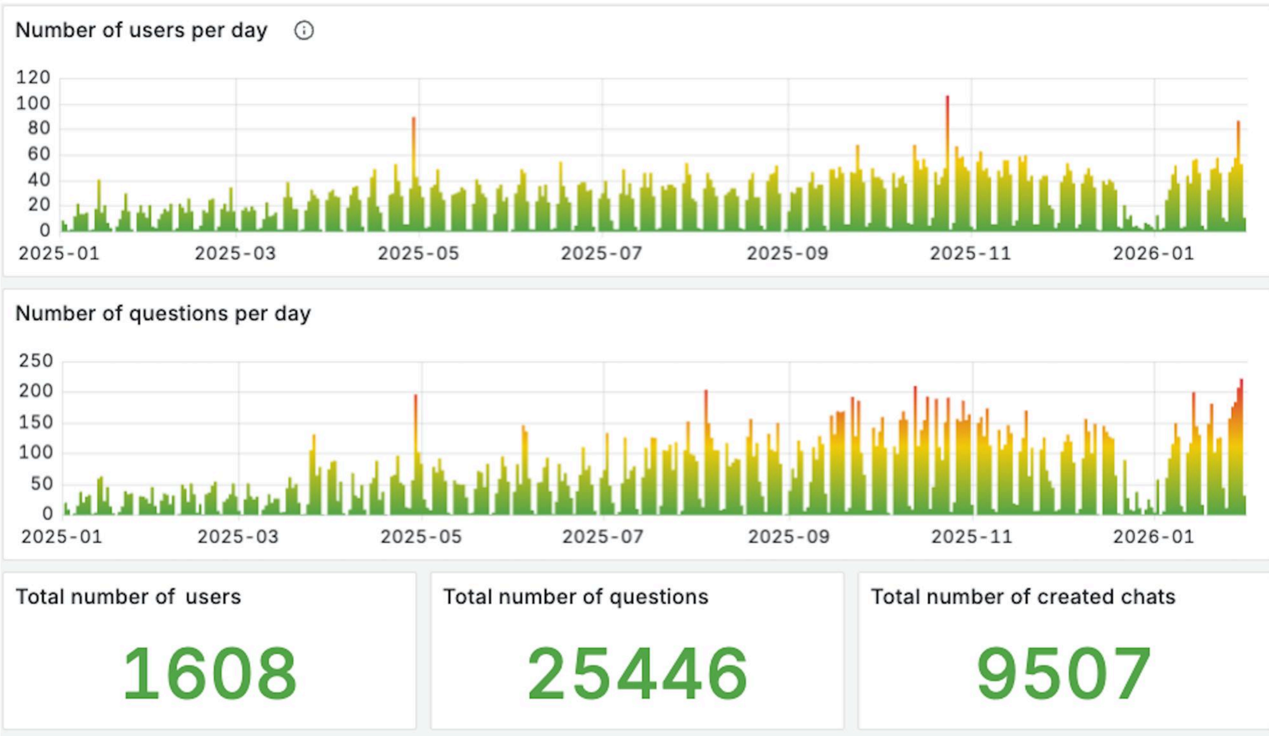


Figure 6. User metrics of Med.ai ASK as of January 2026.

accurate and contextually rich responses, often outperforming unsupervised baselines while remaining competitive with instruction-finetuned models such as Med-PALM2 and Med-Gemini. The integration of diverse retrieval tools has been shown to enhance performance, with detailed ablation studies and usage analyses highlighting the nuanced contributions of each tool. Finally, we integrated Med.ai ASK into a chatbot platform. The encouraging feedback from biomedical experts underscores the practical utility of Med.ai ASK and its potential to drive future advancements not only in question answering but also other tasks such as literature review for the biomedical domain.

There are several lines of future work for our agent. Firstly, we can improve the retrieval step by using snippet extraction and ranking.[7] Doing so will reduce noisy information fed into the generation step, hence improving answers' quality. We can also enrich the underlying knowledge bases by integrating web search tools and connecting with our stakeholders' internal tools and databases. Finally, we plan to enhance the memory mechanism by using other methods such as MemGPT[33] and Memory OS.[34]

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Author contributions

Nhung T. H. Nguyen (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Dmytro Lituiev (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Zhimin Liu (Conceptualization, Data curation, Methodology, Software, Supervision, Writing—review & editing), Aditya Kashyap (Conceptualization, Methodology, Software, Writing—review & editing), Garrett Jenkinson (Conceptualization, Methodology, Software, Visualization, Writing—review & editing), Kevin Kuhl (Investigation, Methodology, Software, Visualization, Writing—review & editing), Christopher Corrado (Methodology, Resources, Software, Writing—review & editing), Naisargi M. Patel (Data curation, Investigation, Validation, Writing—review & editing), Kirti Snigdha (Data curation, Investigation, Validation, Writing—review & editing), Sirwe Saeedi (Investigation, Validation, Writing—review & editing), David Smith (Conceptualization, Funding acquisition, Validation, Writing—review & editing), Nicolas Baro (Funding acquisition, Methodology, Software, Supervision, Writing—review & editing), and Timothy Schultz (Conceptualization, Funding acquisition, Methodology, Software, Supervision, Validation, Writing—original draft, Writing—review & editing)

Supplementary material

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Conflicts of interest

None declared.

Data availability

Data are provided within the published article and its online supplementary material. The in-house dataset remains inaccessible to the public because it contains confidential information related to Johnson & Johnson programs.

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Research and Applications